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DYNAMIC MANUSCRIPT

Robot-assisted gastric endoscopic submucosal dissection significantly improves procedure time at challenging dissection locations

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Abstract

Background Endoscopic submucosal dissection (ESD) is the standard treatment for early malignant stomach lesions. However, this procedure is technically demanding and carries a high complication risk. The level of difficulty in performing ESD is influenced by the location of the lesion. In our study, we aimed to investigate and analyze the effectiveness of robot-assisted ESD for lesions situated in challenging locations within the stomach.

Methods We developed a gastric simulator that could be used to implement various gastric ESD locations. An EndoGel (Sunarrow, Tokyo, Japan) was attached to the simulator for the dissection procedures. Robot-assisted or conventional ESD was performed at challenging or easy locations by two ESD-trainee endoscopists.

Results The procedure time was remarkably shorter for robotic ESD than conventional dissection at challenging locations (6.2 vs. 10.2 min, $P < 0.05$), mainly due to faster dissection (220.3 vs. 101.9 mm^2/min, $P < 0.05$). The blind dissection rate was significantly lower with robotic ESD than with the conventional method (17.6 vs. 35.2%, $P < 0.05$) at challenging locations.

Conclusion The procedure time was significantly shortened when robot-assisted gastric ESD procedures were performed at challenging locations. Therefore, our robotic device provides simple, effective, and safe multidirectional traction for endoscopic submucosal dissection at challenging locations, thereby reducing difficulty of the procedure.

Keywords Endoscopic submucosal dissection · Robotic dissection · EndoGel · Simulator

Gastric endoscopic submucosal dissection (ESD) is a minimally invasive curative treatment for early gastric cancer and has become increasingly popular worldwide [1]. ESD is technically demanding and has a high risk of complications, thereby making it challenging for beginners [2]. Gaining experience through training under the supervision of an experienced practitioner is ideal; however, this is not feasible for everyone. To date, ESD is taught using the traditional master–apprentice model. Trainees observe and assist ESD procedures; initially thereafter, they perform supervised ESD on patients with small lesions on the distal site, followed by progression to increasingly difficult cases [3, 4]. Due to the scarcity of endoscopy centers providing such training, trainees often visit renowned centers located at a distance or practice with an ex vivo/in vivo stomach model [5]. Therefore, it is necessary to develop a device that can assist unskilled individuals in performing high-level ESD with minimal learning curve [6, 7]. Considering these points, we developed a robot-assisted traction device for therapeutic endoscopy.

In the process of gaining expertise in ESD, trainees often encounter difficulties while operating on certain locations of the stomach. Operating on lesions where it is hard to identify the submucosal dissection plane, where endoscopic maneuvers are restricted, and where complications occur frequently, presents significant challenges. According to literature, operating on the upper and middle thirds of the stomach is known to be more time-consuming and often

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cause more complications than the lower third [8, 9], which can be attributed to the fact that in these circumstances, the ESD is generally conducted with the endoscope in a retroflexed or unstable position. In our previous study, we successfully performed ESDs on porcine stomachs in vivo using assistive robotic devices [10] and proved that dissection speed and safety can be improved for both unexperienced and experienced endoscopists by using robotic devices. We hypothesized that there would be greater improvements in submucosal dissection speed and safety with the use of our robotic endoscope in challenging stomach locations. We, therefore, developed a simulator that can implement various ESD locations in the stomach. The challenging locations of the stomach were reproduced using this simulator, and ESD was performed using our robotic device. The current study was conducted to determine if there is an improvement in safety and dissection speed compared to conventional ESD especially in challenging locations.

Materials and methods

This prospective comparative study was performed between May 2022 and September 2022 at Korea University Anam Hospital and aimed to investigate the effectiveness of a robotic assistive device for improving submucosal visualization and dissection speed during ESD at challenging locations in the stomach. We expected to shorten the overall procedure time by improving the submucosal dissection speed when using robotic devices. We designed a two-stage experiment to evaluate the efficacy of the robotic device. We first conducted a series of ESDs using our novel ESD simulator to determine challenging locations of the stomach. Robotic and conventional ESD were then performed alternately at challenging and easy locations. Procedures and devices used in this research do not require ethical approval.

Selection of challenging dissection locations using novel automated gastric simulator

We developed a novel gastric ESD training simulator (Fig. 1) comprising a setting frame with EndoGel™ (Sunarrow Limited, Tokyo, Japan) attached to it. This gastric ESD training model can accurately simulate the location of the human stomach in non-animal subjects. Various stomach locations can be implemented by adjusting the angle of the setting frame and standardizing the process. This model could be set up at 8 different locations, including two vertical locations (middle to upper body, and antrum) and four circumferential locations (anterior wall, posterior wall, lesser curvature, and greater curvature). Considering that the mucosal surface of the stomach approaches the endoscope during actual air suction, a function to simulate air suction was installed. On switching on the air suction button on the simulator controller, the simulator body with the EndoGel moves closer to the endoscope, mimicking air suction. For an objective evaluation of the training model, two independent ESD experts who did not have prior involvement in its development were engaged. They individually evaluated the simulator while performing conventional ESD across eight different gastric locations (Fig. 1). Four locations implemented lesions on antrum (lesser curvature, greater curvature, anterior wall, posterior wall), while the other four implemented lesions on middle to upper body (lesser curvature, greater curvature, anterior wall, posterior wall). Markings of a virtual lesion with a diameter of 20 mm were made on EndoGel with an ESD dual knife (KD-650Q; Olympus, Tokyo, Japan). Conventional ESDs were performed twice on each lesion. The participants graded difficulty of each procedure upon its completion, using the National Aeronautical and Space Administration (NASA) Task Load Index (TLX) which is a quantitative scoring system developed and validated by NASA to evaluate the workload in a procedure. Based on these scoring results and total procedure time results, the most challenging and easiest locations for gastric ESD were selected and the robotic experiments were planned at those locations.

Assistive robotic traction devices

The robotic endoscopic surgical system (ROBOPERA, ENDOROBOTICS, Seoul, Korea) provides multidirectional traction during ESD. It consists of a robotic manipulator with disposable cartridge, two types of intuitive controllers, and a driver unit (Fig. 2). The robotic manipulator is attached to the distal end of the conventional endoscope; it has wrist and elbow joint, and can grasp the mucosa, lift up or pull down, and move circumferentially along the tip of the endoscope. The endoscopist operated the robotic manipulator by controlling the controller attached to the control body of the endoscope. As the controller has the same structure and degrees of freedom as the robotic arm, it allows the robotic arm to be intuitively controlled. The ROBOPERA can be used simultaneously with commercial endoscopic therapeutic instruments, such as the electric dual knife. The robotic arm grasps and lifts the mucosa, thereby exposing dissection plane. ESD was performed according to the manufacturer's protocol.

ESD procedures

The main ESD procedures were performed by two endoscopists who are under the advanced endoscopy fellowship program of our hospital. They had performed over 30, but less than 100 human gastric ESDs. Each endoscopist performed 12 ESDs each at easy and challenging locations.

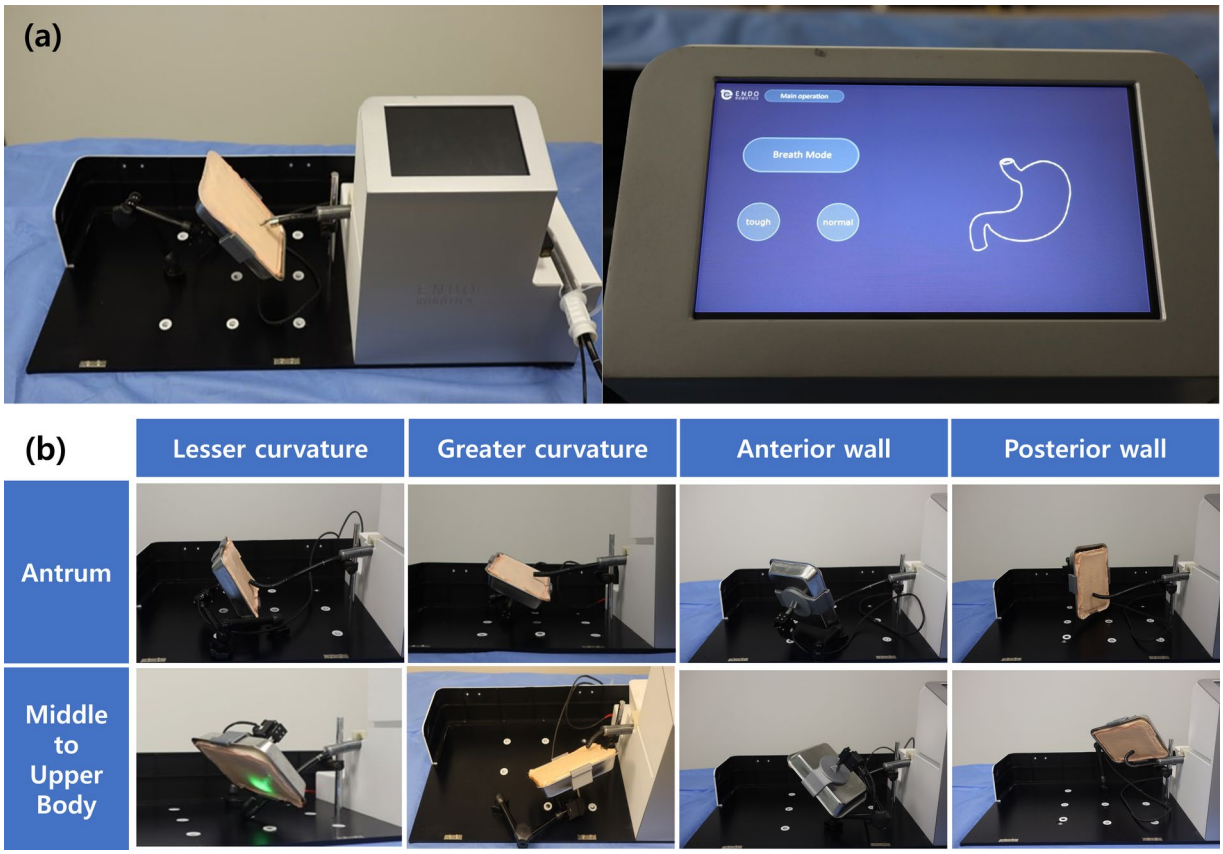

Fig. 1 **a** Overview of the gastric ESD training simulator. Pressing "Breathing mode" on the display, the simulator moves up and down in a shallow motion. **b** Set-up of eight gastric locations. The first row is the antrum. From left, lesser curvature, greater curvature, anterior wall, posterior wall. The second row is the middle to upper body. From left, lesser curvature, greater curvature, anterior wall, posterior wall

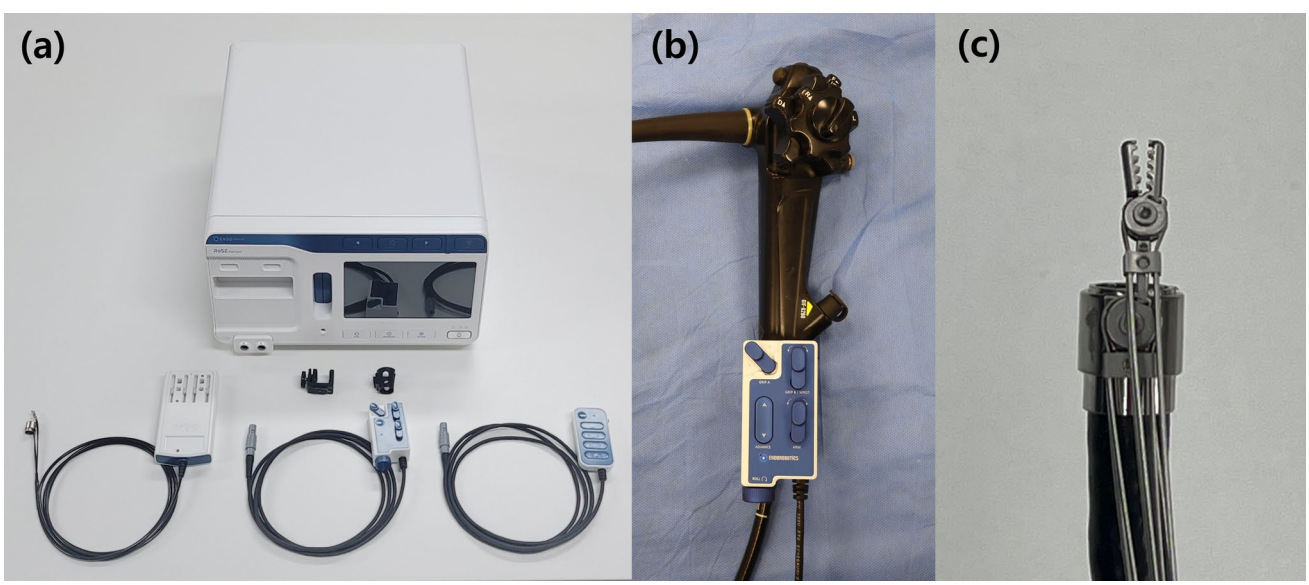

Fig. 2 Overview of the robotic traction device. **a** The robotic endoscopic surgical system (ROBOPERA, ENDOROBOTICS, Seoul, Korea) consists of a robotic manipulator with disposable cartridge, two types of intuitive controllers, and a driver unit. **b** The controller is attached to the control body of the endoscope. **c** The robotic manipulator is attached to the distal end of the conventional endoscope

They alternately performed conventional ESD (CESD) and robot-assisted ESD (RESD) for each lesion. The ESD procedure was standardized according to the following protocol. Lesions measuring 25 mm in diameter were marked with a pen on the EndoGel model. CESDs and RESDs were performed as follows: a transparent cap (FM-EC0007; Finemedix, Seoul, Korea) or robotic manipulator was attached to the distal end of a standard endoscope (GIF-Q260; Olympus Inc., Tokyo, Japan) and then moved toward the target lesion. The lesion was circumferentially marked with a dual knife (KD-650Q; Olympus, Tokyo, Japan). Circumferential mucosal incision and submucosal dissection were performed using the same electric dual knife. In the RESD group, conventional submucosal dissection was conducted until adequate space was available for grasping the mucosa. The dissected mucosal flap was then grasped using the robotic device to expose the submucosa, and submucosal dissection was carried out. Upon completion, the lesion was retrieved and the size of the specimen was measured after being pinned to a corkboard with adequate stretching. The procedures were recorded and reviewed by two independent assessors. Like in the first stage experiment, NASA-TLX score was graded by the participants after each experiment.

Outcome measurements and statistical analyses

The following parameters were recorded for all ESD procedures: location, total procedure time, operative time for specific ESD steps (such as dissection, incision, and grasping), size of the resected tissue, en bloc or piecemeal resection, and adverse events. The primary outcome measure was the total procedure time, which was calculated as the time from the initial mucosal marking to complete resection of the lesions. The secondary outcome measures included submucosal dissection speed (mm^2/min), blind dissection rate, en bloc resection, and complications. The speed of submucosal dissection was determined by dividing the resected area of the specimen divided by the dissection time. A blind dissection was defined as a submucosal dissection in which the dissection plane was not visible. The blind dissection rate is the ratio of time spent on blind dissection to the total dissection time. The grasping time was defined as the duration from when the robotic arm was extended to when submucosal dissection begins after holding the flap properly. To ensure accuracy, the recorded video was analyzed using a stopwatch. Time periods were calculated as the mean time reported by two independent assessors after reviewing each procedure on video. The remaining surface of the EndoGel after the incision was examined to verify muscle injury or perforation. Continuous variables are reported as mean ± standard deviation, and categorical variables are reported as numbers. Ordinal qualitative and quantitative variables were compared using the Wilcoxon or Kruskal–Wallis rank-sum test. Paired comparisons of qualitative variables were performed using Fisher's exact test or the χ^2 tests. Statistical significance was set at $P < 0.05$, and the results were considered statistically significant. SPSS software (version 25.0; IBM Corp., Armonk, NY, USA) was used for all the analyses.

Results

Challenging ESD locations

A total of 32 ESDs were conducted using a simulator by two experts for ESD lesion selection. The lesions with the highest and lowest NASA-TLX scores were the lesser curvature of the middle to upper body and the lesser curvature of the antrum (52.8 vs. 30.2), respectively (Table 1). The procedure time was compared between each location. The procedure time was highest on the lesser curvature of the middle to upper body and lowest on the lesser curvature of the antrum (12.6 vs. 6.1 min). We determined the lesser curvature of the middle to upper body as a challenging lesion and the lesser curvature of the antrum as an easy lesion. We performed CESD and RESD on these two lesions and compared the results.

Robot-assisted ESD significantly improves procedure time in challenging locations

A total of 48 lesions were successfully removed by two trainee endoscopists using either the RESD ($n = 24$) or CESD method ($n = 24$). A total of 24 ESD procedures (12 RESDs and 12 CESDs) each were performed on the lesser curvature of the middle to upper body of the stomach (Fig. 3; Video 1). Another 24 ESD procedures (12 RESDs and 12 CESDs) were performed on the lesser curvature of the antrum. In the "challenging location", the lesser curvature

Table 1 Comparison of average NASA-TLX scores and total procedure time by ESD location

Outcomes		NASA TLX score	Total procedure time (min)
Antrum ($n = 16$)	Lesser curvature ($n = 4$)	30.2	6.1 ± 3.4
	Greater curvature ($n = 4$)	40.6	7.6 ± 4.2
	Anterior wall ($n = 4$)	37.3	8.4 ± 3.2
	Posterior wall ($n = 4$)	43.6	8.0 ± 2.7
Middle to upper body ($n = 16$)	Lesser curvature ($n = 4$)	52.8	12.6 ± 4.1
	Greater curvature ($n = 4$)	42.2	10.4 ± 4.2
	Anterior wall ($n = 4$)	46.4	10.3 ± 5.1
	Posterior wall ($n = 4$)	45.4	9.8 ± 3.8

Values are mean or mean ± standard deviation

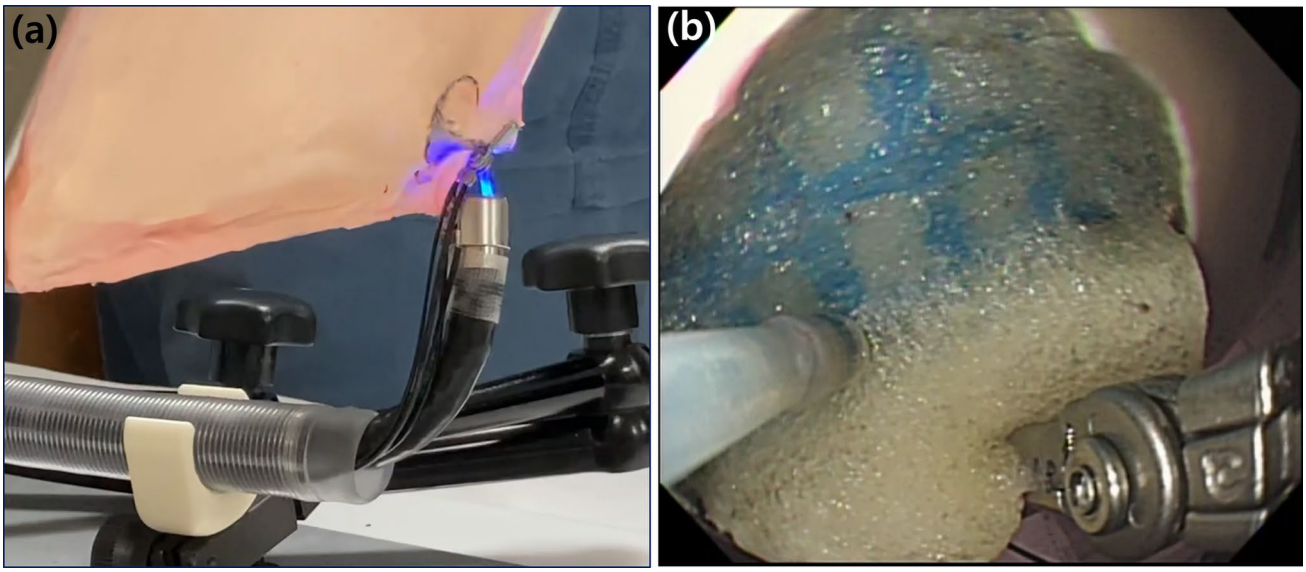

Fig. 3 Robot-assisted endoscopic submucosal dissection at a challenging location. **a** The dissection plane was easily exposed in robotic ESD at challenging location. **b** Endoscopic view of robotic ESD

Table 2 Outcomes of endoscopic submucosal dissection at challenging and easy locations

Outcomes	R-ESD ($n=24$)	Conventional ESD ($n=24$)	*P* value
Lesser curvature side of middle to upper body (challenging location)			
Dissection area, mm^2	710.4 ± 13.4	735.1 ± 19.1	0.30
En bloc resection rate, %	100	100	NA
Total procedure time, min	6.2 ± 0.4	10.2 ± 0.9	< 0.05
Dissection time, min	3.4 ± 0.2	8.3 ± 0.8	< 0.05
Dissection speed (mm^2/min)	220.3 ± 21.2	101.9 ± 14.9	< 0.05
Grasping time (min)	0.8 ± 0.1	NA	NA
Perforations during the procedure, %	0	0	NA
Muscle injuries during the procedure, %	0	12%	NA
Blind dissection rate, %	17.6 ± 4.6	35.2 ± 0.9	< 0.05
NASA TLX score	35.2 ± 4.8	44.6 ± 6.4	< 0.05
Lesser curvature side of antrum (easy location)			
Dissection area, mm^2	788.7 ± 20.4	736.2 ± 25.8	0.12
En bloc resection rate, %	100	100	NA
Total procedure time, min	6.8 ± 0.3	7.4 ± 0.7	0.58
Dissection time, min	4.2 ± 0.1	4.9 ± 0.2	< 0.05
Dissection speed (mm^2/min)	193.2 ± 11.5	153.6 ± 8.4	< 0.05
Grasping time (min)	0.4 ± 0.1	NA	NA
Perforations during the procedure, %	0	0	NA
Muscle injuries during the procedure, %	0	0	NA
Blind dissection rate, %	13.6 ± 6.6	24.2 ± 0.7	< 0.05
NASA TLX score	36.2 ± 6.0	38.6 ± 5.4	0.24

Values are *n*/*N* (%) or mean ± standard deviation

ESD endoscopic submucosal dissection, *RESD* robotic-assisted ESD

side of middle to upper body, the procedure time was significantly shorter for RESD than CESD (6.2 min vs. 10.2 min, $P<0.05$) (Table 2). The submucosal dissection speed of RESD was significantly faster than that of CESD (220.3 vs. 101.9 mm^2/min, $P<0.05$). In the RESD group, the average blind dissection rate was 17.6%, compared to 35.2% in the CESD ($P<0.05$) group. During RESD, it took an average of 48 s to grasp the flap using a robotic arm after making the flap. There were no muscle injuries in the RESD group. However, three cases of muscle injuries occurred in the CESD group. The total NASA-TLX score was significantly lower in the RESD group compared to the CESD group (35.2 ± 4.8 vs 44.6 ± 6.4, $P<0.05$).

On the other hand, the total procedure time in the RESD group was shorter than that in the CESD group on the lesser curvature of the antrum, but without significance (6.8 vs. 7.4 min, $P=0.58$). The submucosal dissection speed of the RESD group was significantly faster than that of the CESD group (193.2 vs. 153.6 mm^2/min, $P<0.05$). However, the difference was less than the difference in challenging locations. There were no complications, such as muscle injuries or perforations in either group. There was no significant difference in the NASA-TLX total score between the CESD and RESD groups.

In the CESD group, ESD performed at challenging location resulted in significantly slower dissection speed than those performed in easy locations. (101.9 vs. 153.6 mm^2/min, $P<0.05$). For RESD, there was no significant difference in dissection speed when comparing different locations (220.3 vs. 193.2 mm^2/min, $P=0.27$).

Discussion

In general, ESD is taught using the traditional master–apprentice model. The trainees observe and assist in ESD procedures, followed by performing ESD on patients under supervision, starting from small lesions in the distal stomach and progressing to increasingly difficult cases [4]. The usefulness of ESD has already been established. However, ESD education is yet to overcome various obstacles, including a flat learning curve and lack of educational resources [11]. ESD trainees visit large volume centers to learn ESD or practice ESD using animal models [3]. Both methods are economically challenging and have ethical issues, especially experimentation on animals. Therefore, it is essential to develop ESD simulators and assistive devices that can aid in the practice of ESD alone. We developed an ESD simulator that ensures repetitive practice for trainees by reproduction of different ESD locations, without supervision of an experienced endoscopist.

Various factors, including lesion characteristics (such as vascularity, ulceration, scarring, and fibrosis) and technical considerations (such as device angle and scope maneuverability), determine ESD difficulty [9]. The optimal position in ESD for beginners is the gastric antral lesion [12], which is easily accessible with a favorable wall thickness, and allows a stable endoscope position for forward viewing and retroflexion. In terms of ESD difficulty based on location, the longest procedure time has been reported for the upper third of stomach [8, 9, 13]. Compared to the lower third lesions, the upper third lesions allow an unstable endoscopic position and are more likely to undergo ESD in the retroflex position. Accurate manipulation of the endoscope in the retroflex position is particularly important while performing gastric ESD for the middle to upper-body lesser curvature. Dissection while positioning the endoscope in an "inverted" fashion requires intense practice and training. Traditionally, case numbers have been used as a surrogate for determining the competency in ESD. In literature, the number of cases required to achieve competency varies between 30 and 80 [12, 14].

Experts mainly recommend gaining experience with antral lesions and challenging upper-third lesions [14]. We attempted to reproduce stomach positions using our novel ESD simulator. The experts used the simulator to evaluate similarity between the reproduced locations and the actual stomach position, and the ESD difficulty based on the location. All locations were evaluated as being similar to the actual gastric locations. The procedure time was longer in the lesser curvature of the middle to upper portion than at other locations, similar to the previous reports on actual stomach ESD [15, 16]. The robotic ESD traction device that we developed is particularly suitable for lesions that present technical challenges for ESD [10]. We compared the performance of RESD and CESD methods by reproducing a challenging ESD location in the simulator.

Two endoscopists with indeterminate experience, who did not have familiarity with difficult positions of ESD, participated in the experiment. Visualization of the submucosal dissection plane in the retroflex position was difficult in CESD. We attempted to expose the submucosal dissection plane by using gravity, which was not technically feasible. A clear difference was observed when performing ESD at a challenging location using the robotic device. The flap was held accurately by the robotic arm and the dissection plane was easily exposed in all cases (Fig. 4). In most cases, the dissection could be completed by holding the flap at once without the need to correct the flap again. In six cases, as the dissection progressed, the flap was corrected once more for better visualization. A significant difference in the procedural time and dissection speed between RESD and CESD was observed in challenging locations compared to easy locations. In the study, the RESD group demonstrated a notably lower rate of blind dissections and a more favorable safety profile compared to the CESD group. Our

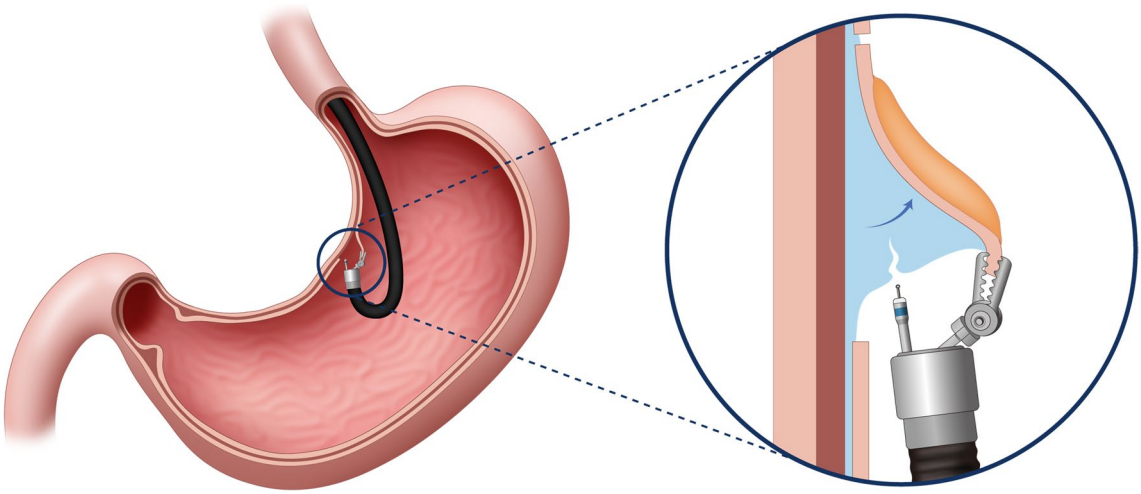

Fig. 4 Graphical illustration of gastric ESD procedure using our robotic device. Multidirectional traction with robotic arm provides a clear visualization of the submucosal plane, even in challenging locations

robotic device enables fast and safe ESD regardless of the ESD location, minimizing the impact of location on ESD. When beginners perform ESD in challenging lesions, utilizing a robotic assistive device appears to reduce the learning curve, allowing for a quicker and safer execution of the ESD procedure.

This study has some limitations. First, it only considered a few factors influencing the difficulty of ESD. The factors affecting ESD difficulty are diverse and interact with each other [15, 17]. However, it is technically difficult to simulate blood vessels, degree of fibrosis, stomach wall thickness, and location of water accumulation. Instead, to maximize training effectiveness, we added a "breathing mode" to the simulator options that simulates patient breathing. When "breathing mode" is enabled, the simulator moves up and down in a shallow motion, just like a patient breathing. Second, a relatively small number of endoscopists from a single center participated in this study. If endoscopists with more diverse proficiency levels had been included, more useful results could have been obtained. When performing ESD in humans, various auxiliary methods, such as air suction or patient position changes, can be considered. However, simulators may not accurately implement these aspects [18]. Moreover, the assessment of challenging locations relied on subjective measures through a questionnaire. Hence, we conducted an analysis of the procedural time for each location to establish an objective evaluation.

Conclusion

The procedure time improved significantly when assistive robots aided the gastric ESD procedures in challenging locations. Our novel gastric ESD simulator demonstrated highly reproducibility for each gastric location and could potentially be used for ESD training. Overall, our robotic device offers easy-to-use, effective, and safe multidirectional traction during gastric ESD in challenging locations.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s00464-024-10743-9.

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Declarations

Disclosures Hyuk Soon Choi, Bora Keum, and Hoon Jai Chun hold stocks for ENDOROBOTICS. Joo Ha Hwang is a consultant for Olympus, Medtronic, Boston Scientific, Micro-Tech, LumenDi, Steris and ENDOROBOTICS. Sang Hyun Kim, Taebin Kwon, Chanwoo Kim, Seonghyeon Won, Han Jo Jeon, Eun Sun Kim, and Yoon Tae Jeen have no conflicts of interest or financial ties to disclose.

Ethical approval This research does not require ethical approval.

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